FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
FULL TEST – X
PAPER-1

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**A, B, C, D**	19.	**A, B, C, D**	37.	**A, B, C**
2.	**B, C, D**	20.	**C, D**	38.	**A, C**
3.	**A, B**	21.	**A, B, C**	39.	**C, D**
4.	**B, C**	22.	**B**	40.	**A, B**
5.	**A, C, D**	23.	**A, B**	41.	**D**
6.	**A, B**	24.	**B**	42.	**B, D**
7.	**A, B, C**	25.	**B, C**	43.	**A, B, C, D**
8.	**C**	26.	**B**	44.	**C**
9.	**B**	27.	**C**	45.	**D**
10.	**C**	28.	**D**	46.	**D**
11.	**D**	29.	**C**	47.	**A**
12.	**B**	30.	**D**	48.	**C**
13.	**C**	31.	**B**	49.	**D**
14.	**9**	32.	**9**	50.	**4**
15.	**4**	33.	**7**	51.	**7**
16.	**3**	34.	**3**	52.	**1**
17.	**5**	35.	**6**	53.	**3**
18.	**5**	36.	**7**	54.	**9**

Physics

PART – I

SECTION – A

1. Time period of oscillation $T = 2\pi\sqrt{\frac{\text{reduced mass}}{k}} = 2\pi\sqrt{\frac{2m}{3k}}$

Required time $t = \frac{T}{4} = \frac{\pi}{2}\sqrt{\frac{(m\times 2m)}{3mk}} = \frac{\pi}{2}\sqrt{\frac{2m}{3k}}$

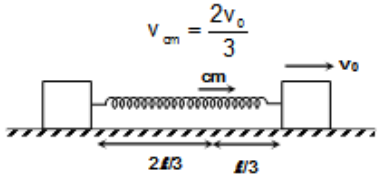
$V_{cm} = \frac{2mv_0}{3m} = \frac{2v_0}{3}$

Initially at mean position the block of mass 2m is moving with velocity $\frac{v_0}{3}$ towards right w.r.t centre of mass. Hence its minimum speed is $\left|\frac{v_0}{3} - \frac{2v_0}{3}\right| = \frac{v_0}{3}$

In centre of mass frame for block of mass 2m,

$\frac{1}{2}(2m)\frac{v_0^2}{9} = \frac{1}{2}(3k)A^2$

$\Rightarrow \quad A = \sqrt{\frac{2mv_0^2}{27k}}$



2. Let $M = G^a\eta^bB^C$

$\Rightarrow \quad M = \left[M^{-1}L^3T^{-2}\right]^a\left[ML^{-1}T^{-1}\right]^b\left[ML^{-1}T^{-2}\right]^c$

$\Rightarrow \quad a = \frac{1}{2}, b = 4, c = -\frac{5}{2}$

Further Let $L = G^a\eta^bB^c$

$= \left[M^{-1}L^3T^{-2}\right]^a\left[ML^{-1}T^{-1}\right]^b\left[ML^{-1}T^{-2}\right]^c$

$\Rightarrow \quad a = \frac{1}{2}, b = 2, c = -\frac{3}{2}$

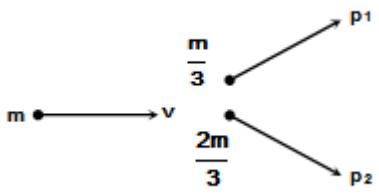
3. $p_1^2 + p_2^2 + 2p_1p_2\cos 60° = p^2$ (where p = mv)

$\Rightarrow \quad p_1^2 + p_2^2 + p_1p_2 = p^2$ …(i)

$\frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1} = \frac{2}{1} \Rightarrow p_1 = 2p_2$ …(ii)

From equation (I) & (II)

$p_2 = \frac{p}{\sqrt{7}} = \frac{mv}{\sqrt{7}}$ and $p_1 = \frac{2p}{\sqrt{7}} = \frac{2mv}{\sqrt{7}}$



$v_1 = \frac{p_1}{m/3} = \frac{6v}{\sqrt{7}}$ and $v_2 = \frac{p_2}{2m/3} = \frac{3v}{2\sqrt{7}}$

Increase in kinetic energy of the system due to explosion

$$= \frac{1}{2}\frac{m}{3}\left(\frac{6v}{\sqrt{7}}\right)^2 + \frac{1}{2}\left(\frac{2m}{3}\right)\left(\frac{3v}{2\sqrt{7}}\right)^2 - \frac{1}{2}mv^2 = \frac{13}{28}mv^2$$

4. Magnetic flux in a circular loop of radius r

$$= \int_0^r B(2\pi r)dr = \int_0^r \frac{Kt}{r}2\pi r dr = 2\pi Ktr$$

$$\oint \vec{E}.\vec{dr} = \frac{-d\phi}{dt}$$

$$\Rightarrow \quad E(2\pi r) = -2\pi Kr$$

$$\Rightarrow \quad E = -K$$

Velocity of the bead $V = at = \frac{qE}{m}t = \frac{Kqt}{m}$

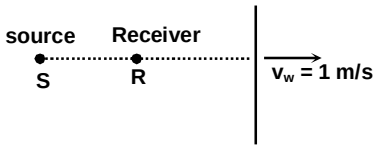
Further

$$qvB + N = \frac{mv^2}{R}$$

$$\Rightarrow \quad \frac{K^2q^2t^2}{mR} + N = \frac{m}{R}\frac{K^2q^2t^2}{m^2}$$

$$\Rightarrow \quad N = 0$$

5. $f_w = \frac{v - v_w}{v}f$



$$f_1 = \frac{v}{v + v_w}f_w = \frac{v}{v + v_w}\left(\frac{v - v_w}{v}f\right)$$

$$f_1 = \frac{v - v_w}{v + v_w}f$$

$$\lambda_1 = \frac{v}{f_1} = \frac{v + v_w}{(v - v_w)}\lambda = \frac{1 + \frac{v_w}{v}}{1 - \frac{v_w}{v}}\lambda$$

$$= \left(1 + \frac{2v_w}{v}\right)\lambda$$

% change in wavelength = $\frac{\lambda_1 - \lambda}{\lambda} \times 100 = \frac{2v_w}{v} \times 100 = \frac{200}{1000/3} \simeq 0.6\%$

Frequency of the echo heard by the observer = $f_1 = \frac{v - v_w}{v + v_w}f \simeq \left(1 - \frac{2v_w}{v}\right) \times 1000$

$$= \left(1 - \frac{2}{1000/3}\right) \times 1000 = 994 \text{ Hz}$$

Beat frequency detected by the receiver = $f - f_1 \simeq 6\text{Hz}$

6. From conservation of momentum

$$\vec{p}_\alpha + \vec{p}_d = 0$$

$$\Rightarrow \quad \vec{p}_d = -\vec{p}_\alpha \Rightarrow p_d = p_\alpha = \sqrt{2m_\alpha k_\alpha}$$

$$v_d = \frac{p_d}{m_d} = \sqrt{\frac{2m_\alpha k_\alpha}{m_d}} = \frac{\sqrt{2 \times 4 \times 1.66 \times 10^{-27} \times 5.77 \times 1.6 \times 10^{-13}}}{202 \times 1.66 \times 10^{-27}} \approx 3.4 \times 10^5 \text{ m/s}$$

Energy released $= \frac{m_p}{m_d} k_\alpha = \frac{206}{202} \times 5.77 = 5.88$ MeV

7. When the switch S was open, charge on the capacitors, $q_1 = C_{eq} v = \frac{4}{3} \times 3 = 4\mu C$. When the switch S is closed then in steady state charge on capacitor of capacitance $2\mu F$ will be zero and charge on 4 μF capacitor $q_2 = 4 \times 3 = 12\mu C$. After the switch S was closed.

Energy supplied by the battery = $\varepsilon \Delta q = 3 \times 8 = 24\ \mu J$

Change in energy stored in capacitors = $\frac{1}{2} 4 \times 9 - \frac{1}{2} \frac{4}{3} \times 9 = 12\ \mu J$

Hence, Heat produced = 24 – 12 = 12 μJ

8-10. (I) $\delta(P_1) = 0$, $\phi(P_1) = 0$, $I(P_1) = I_m$

$$\delta(P_2) = d\frac{d/2}{D} = \frac{d^2}{2D} = \frac{10^{-6}}{2 \times 1.5} = \frac{10^{-6}}{3} = \frac{2}{3}\lambda$$

$$\phi(P_2) = \frac{2\pi}{\lambda}\frac{2}{3}\lambda = \frac{4\pi}{3},\ I(P_2) = I_m \cos^2 \frac{2\pi}{3} = \frac{I_m}{4}$$

$\Rightarrow$ Correct choice for Q. 10 is C

(II) $\delta(P_1) = \frac{d}{D}\frac{d}{8} = \frac{d^2}{8D} = \frac{\lambda}{6},\ \ \phi(p_1) = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \frac{\pi}{3}$

$$I(P_1) = I_m \cos^2 \frac{\pi}{6} = \frac{3I_m}{4}$$

$$\delta(P_2) = d\frac{d}{2d} - \frac{d}{D}\frac{d}{8} = \frac{2\lambda}{3} - \frac{\lambda}{6} = \frac{3\lambda}{6} = \frac{\lambda}{2}$$

$I(P_2) = 0$

$\Rightarrow$ Correct choice for Q.9. is B

(III) $\delta(P_1) = (\mu - 1)t = \frac{5\lambda}{12}, \phi(P_1) = \frac{2\pi}{\lambda}\frac{5\lambda}{12} = \frac{5\pi}{6}$

$$I(P_1) = I_m \cos^2 \frac{5\pi}{12}$$

$$\delta(P_2) = d\frac{d}{2D} - (\mu - 1)t = \frac{2\lambda}{3} - \frac{5\lambda}{12} = \frac{3\lambda}{12} = \frac{\lambda}{4}$$

$$\phi(p_2) = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2},\ I(P_2) = I_m \cos^2 \frac{\pi}{4} = \frac{I_m}{2}$$

(IV) $\delta(P_1) = d\frac{5d}{8D} + (\mu - 1)t = \frac{5}{4}\frac{2\lambda}{3} + \frac{\lambda}{3} = \frac{5\lambda}{6} + \frac{\lambda}{3} = \frac{7\lambda}{6}$

$$\phi(P_1) = \frac{2\pi}{\lambda}\frac{7\lambda}{6} = \frac{7\pi}{3},\ I(P_1) = I_m \cos^2 \frac{7\pi}{6} = \frac{I_m}{4}$$

$$\delta(P_2) = d\frac{5d}{8D} + d\frac{d}{2D} + (\mu - 1)t$$

$$= \frac{5\lambda}{6} + \frac{2\lambda}{3} + \frac{\lambda}{3} = \frac{11\lambda}{6}$$

$$\phi(P_2) = \frac{2\pi}{\lambda}\frac{11\lambda}{6} = \frac{11\pi}{3},\ I(P_2) = I_m \cos^2 \frac{11\pi}{6} = \frac{I_m}{4}$$

11. $\Delta U = -\int pdv$ is valid when process is adiabatic $P \propto T^{7/2}$ implies process is adiabatic

12. $p \propto V^{-3/2} \Rightarrow pV^{3/2} = \text{constant}$

$$C = C_V + \frac{R}{1-\frac{3}{2}} = \frac{3R}{2} - 2R = -\frac{R}{2}$$

$$p \propto V^{-3/2} \Rightarrow T \propto \frac{1}{\sqrt{V}}$$

Since gas is compressed, its temperature increases

$\Delta Q = nC\Delta T < 0$

13. $\Delta U = \Delta Q - p\Delta V$ is valid for isobaric process

$$U \propto \frac{1}{\rho} \Rightarrow T \propto V \text{ or } \frac{V}{T} = \text{ constant}$$

So, process is isobaric

SECTION – C

14. $i_1 + i_2 = i$

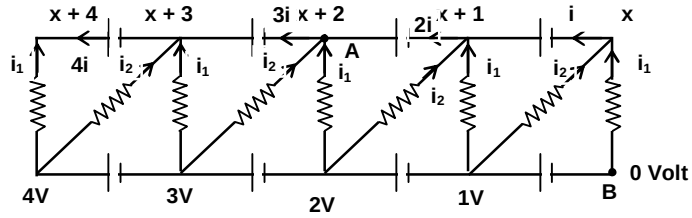
$4i + i_1 = 0$

$\Rightarrow \quad 5i_1 = -4i_2$

$$\Rightarrow \quad \frac{5x}{1} = -4\frac{(x-1)}{1}$$

$$\Rightarrow \quad x = \frac{4}{9}v$$

$$V_A - V_B = x + 2 = \frac{22}{9}V$$



15. $$\tau = -\frac{dU}{d\theta} = -\frac{1}{2}V^2\frac{dc}{d\theta} = -\frac{1}{2}V^2\frac{d}{d\theta}\left\{\frac{\varepsilon_0 R^2\theta}{2d} + \frac{\varepsilon_0 k(\pi-\theta)R^2}{2d}\right\} = \frac{(k-1)\varepsilon_0 R^2 V^2}{4d}$$

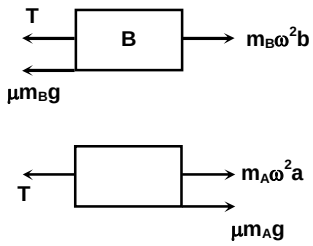
$\therefore \quad n = 4$

16. $T + \mu m_B g = m_B\omega^2 b$

$T = m_A\omega^2 a + \mu m_A g$

$\Rightarrow \quad m_B\omega^2 b - \mu m_B g = m_A\omega^2 a + \mu m_A g$

$$\Rightarrow \quad \omega = \sqrt{\frac{\mu g(m_A + m_B)}{(m_B b - m_A a)}} = \frac{10}{3}\text{ rad/s}$$



17. $\sum F_x = 0$

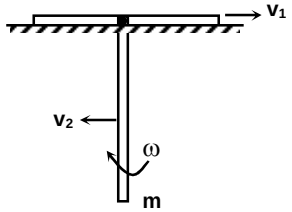
$\Rightarrow \quad mv_1 - mv_2 = 0$

At hinged point

$$v_1 = \frac{L}{2}\omega - v_2$$

From conservation of mechanical energy

$$mgL = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 + \frac{1}{2}\frac{mL^2}{12}\omega^2$$



Solving $v_1 = \sqrt{\frac{3}{5}gL} = 5\,m/s$

18. $-\frac{dN_A}{dt} = \lambda N_A \Rightarrow N_A = N_0 e^{-\lambda t}$

$\frac{dN_B}{dt} = \lambda N_A - 2\lambda N_B$

$\Rightarrow \frac{dN_B}{dt} = \lambda N_0 e^{-\lambda t} - 2\lambda N_B$ …(1)

Solving equation (1)

$N_B = N_0\left(e^{-\lambda t} - e^{-2\lambda t}\right)$

N_B will be maximum, when $\frac{dN_B}{dt} = 0$

$\Rightarrow e^{-\lambda t} = \frac{1}{2}$

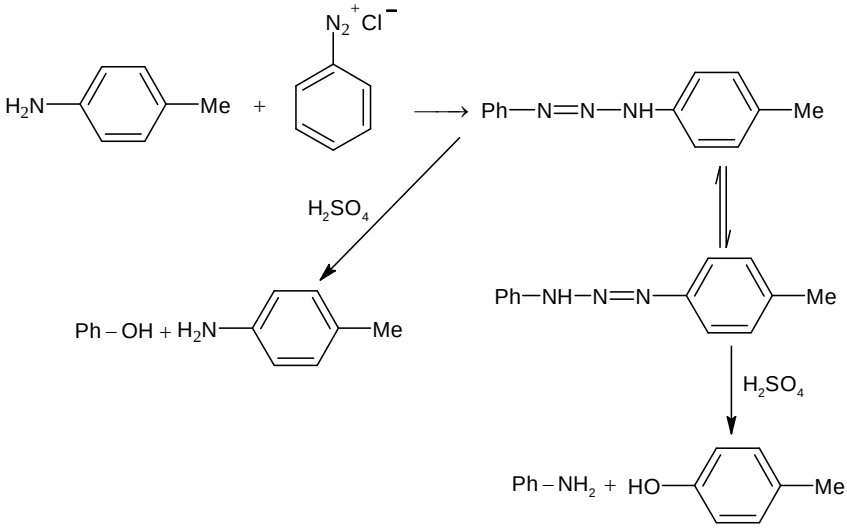
$(N_B)_{max} = N_0\left(\frac{1}{2} - \frac{1}{4}\right) = 5 \times 10^{20}$

Chemistry

PART – II

SECTION – A

19.



20. $Cu^{+2} \rightarrow$ Green colour
$Co^{+3} \rightarrow$ Deep blue.

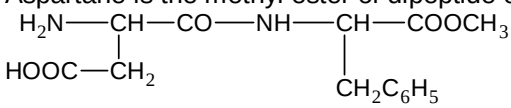
21. Fact

22. $ClF_3 \rightarrow$ Planar distorted – T- shape and polar
$XeF_5^- \rightarrow$ Pentagonal planar and non-polar
$PCl_5 \rightarrow$ Non-planar and non-polar
$BrF_5 \rightarrow$ Non-planar and polar

23. In brown ring test $[Fe(H_2O)_5NO]^{+2}$ complex is formed. Which is of brown colour.
While $Na_2[Fe(CN)_5NO].2H_2O$ is known as sodium nitroprusside where the oxidation number of iron is +1.

24. $CuSO_4$ reacts with KI to give white ppt. of Cu_2I_2 but $CuSO_4$ does not react with KCl to give $CuCl_2$.

25. Monomer of orlon is acrylonitrile
$\Rightarrow CH_2 = CHCN$
Aspartane is the methyl ester of dipeptide of L-aspartic and L-phenylalanine.



26. Fact

27. Fact

28. Fact

29. Fact

30. Fact

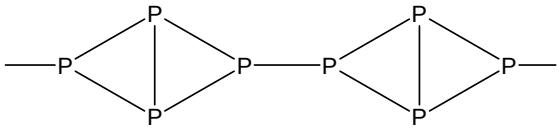
31. Fact

SECTION – C

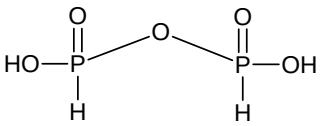
32. NiS and CoS $\rightarrow$ black in colour.
A = 2
B = 4
C = 0
D = 3

33. $A \rightarrow CO_2$
$B \rightarrow NH_3$
$C \rightarrow H_2$
$D \rightarrow CH_3 - CH_3$
$E \rightarrow H_2$
$(y + z + w + v) - x \Rightarrow (17 + 2 + 30 + 2) - 44$
$= 7$

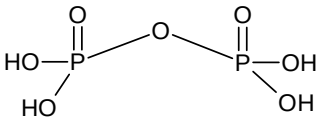
34. (i)



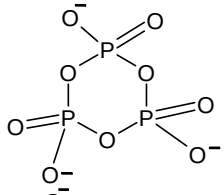
(ii)



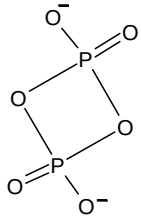
(iii)



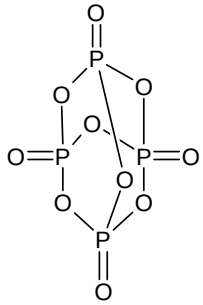
(iv)

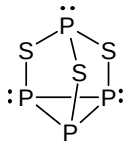


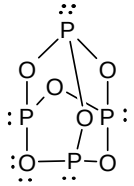
(v)

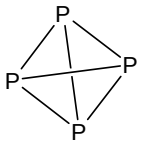


(vi)

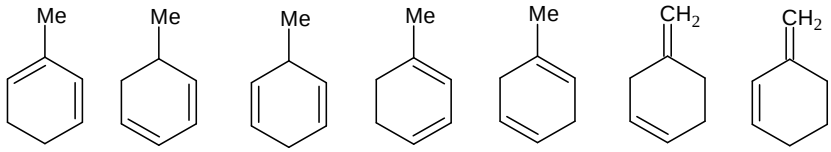


(vii) 

(viii) 

(ix) 

35. (ii), (iii), (v), (vii), (viii), (ix) are correct statements.

36. 

Mathematics

PART – III

SECTION – A

37. (A) Put 2 cot x = t
(B) Put 2 cosec x = t
(C) Put $t^2 = u$ in B
(D) Put cot x + cosec x = e^t

38. $t_n = (n+1)^{1/3} - (n)^{1/3}$
$S_n = \sqrt[3]{2197} - 1 = 12$

39. $\cot 65^\circ + 2\tan 40^\circ = \dfrac{\sin 25^\circ}{\cos 25^\circ} + \dfrac{\sin 40^\circ}{\cos 40^\circ} + \tan 40^\circ$

$= \dfrac{\sin 65^\circ}{\cos 25^\circ \cdot \cos 40^\circ} + \dfrac{\sin 40^\circ}{\cos 40^\circ} = \dfrac{1+\sin 40^\circ}{\cos 40^\circ} = \dfrac{1+\cos 50^\circ}{\sin 50^\circ} = \cot 25^\circ$

and $\sqrt{3}\sin 80^\circ - 2\sin 50^\circ = 2\sin 60^\circ \cdot \sin 80^\circ - 2\sin 50^\circ$
$= \cos 20^\circ - \cos 40^\circ = \sin 10^\circ$

40. $\because$ f is positive, strictly increasing and concave up
$\therefore \lim\limits_{x\to\infty} f(x) = \infty$

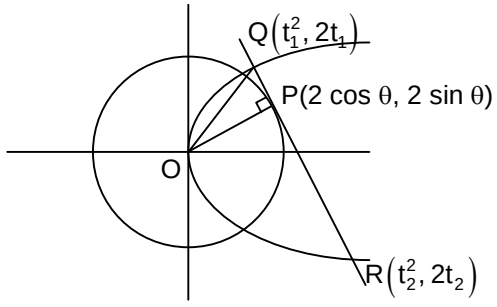
$$\lim_{x\to\infty} \frac{f'(x)}{xf''(x)} = \lim_{x\to\infty} \frac{\frac{f'(x)}{f''(x)}}{x} = \lim_{x\to\infty} \frac{f''(x)f''(x) - f'(x)f'''(x)}{(f''(x))^2}$$ (Applying L' Hospital rule)

$$= \lim_{x\to\infty} \left(1 - \frac{f'(x)f'''(x)}{(f''(x))^2}\right) = \frac{2}{3}$$

$$\lim_{x\to\infty} \frac{xf'(x)}{f(x)} = \lim_{x\to\infty} \frac{f'(x)+xf''(x)}{f'(x)} = \lim_{x\to\infty} \left(1 + \frac{xf''(x)}{f'(x)}\right) = \frac{5}{2}$$ (Applying L' Hospital rule)

$$\lim_{x\to\infty} \frac{f(x)f''(x)}{(f'(x))^2} = \lim_{x\to\infty} \frac{f(x)}{xf'(x)} \cdot \lim_{x\to\infty} \frac{xf''(x)}{f'(x)} = \frac{3}{5}$$

41. $OQ^2 = OP^2 + PQ^2$
$\Rightarrow t_1^2 + 4t_1^2 = 5$
$\Rightarrow t_1 = 1$ ($\because t_1 > 0$)
$\Rightarrow Q \equiv (1, 2)$
$PQ = 1 \Rightarrow \cos\theta + 2\sin\theta = 2$
$\Rightarrow \cos\theta = \dfrac{4}{5}$, $\sin\theta = \dfrac{3}{5}$
$\Rightarrow p \equiv \left(\dfrac{8}{5}, \dfrac{6}{5}\right)$



Now R lies on tangent 4x + 3y = 10
$\Rightarrow t_2 = -\dfrac{5}{2}$ ($\because t_2 < 0$)
$\Rightarrow R \equiv \left(\dfrac{25}{4}, -5\right)$ and $S \equiv \left(-\dfrac{5}{2}, -\dfrac{3}{2}\right)$

42. $\vec{d}\cdot\vec{a} = \left[\vec{a}\ \vec{b}\ \vec{c}\right]\cos y$

$\Rightarrow -\vec{d}\cdot\left(\vec{b}+\vec{c}\right) = [a\ b\ c]\cos y$

$\Rightarrow \cos y = -\dfrac{\vec{d}\cdot\left(\vec{b}+\vec{c}\right)}{[a\ b\ c]}$

Also $\sin x = -\dfrac{\vec{d}\cdot\left(\vec{a}+\vec{b}\right)}{[a\ b\ c]}$ and $\dfrac{\vec{d}\cdot\left(\vec{a}+\vec{c}\right)}{[a\ b\ c]} = -2$

$\therefore \sin x + \cos y + 2 = 0$

$\Rightarrow \sin x = -1, \cos y = -1$

Minimum of $x^2 + y^2 = \dfrac{5\pi^2}{4}$

43. Solving given equations, we get $a(x^3 - 6x^2 + 12x - 8) = x^3 - 12x + 16$

$\Rightarrow a(x-2)^3 = (x-2)^2(x+4)$

$\therefore x = 2$ is a repeated root and hence the two curves touches each other at $x = 2\ \forall\ a \in R$

44.-46. (I) $A = A^T$

$\Rightarrow A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ where $a, b, c \in \{0, \pm1, \pm 2\}$

$\therefore$ Number of such matrices in $A = 5 \times 5 \times 5 = 125$

Now $|A| = 0, \Rightarrow b^2 = ac$

b = 0, 9 matrices

b = ±1, 4 matrices

b = ±2, 4 matrices

———————

Total 17 matrices

$|A| \neq 0$, 125 – 17 = 108 matrices

$Tr(A) = 0 \Rightarrow a + c = 0 \Rightarrow c = -a$, total $5 \times 5 = 25$ matrices

$\therefore$ For $Tr(A) \neq 0 \Rightarrow 100$ matrices

(II) $A = -A^T$

$\therefore A = \begin{bmatrix} 0 & a \\ -a & 0 \end{bmatrix}$ $a \in \{0, \pm1, \pm 2\}$

$\Rightarrow$ 5 matrices

$|A| = 0 \Rightarrow a^2 = 0, a = 0 \Rightarrow 1$ matrix

Here $Tr(A) = 0$

(III) Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow A^2 = I$

$\Rightarrow a^2 + bc = d^2 + bc = 1$

$\Rightarrow b(a + d) = c(a + d) = 0$

$Tr(A) = 0 \Rightarrow a + d = 0, bc = 1 - a^2$

a = 0, 2 matrices

a = ±1, 18 matrices

a = ±2, not possible $\Rightarrow$ 20 matrices

$Tr(A) = a + d \neq 0 \Rightarrow b = c = 0$ and $a = d = \pm1$

$\Rightarrow$ 2 matrices

If $A^2 = I$, then $|A| = 1 \neq 0$

(IV) $AA^T = I$

$\Rightarrow$ A is orthogonal matrix

$\Rightarrow$ Rows of A are mutually perpendicular unit vector

$\Rightarrow A = \begin{bmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{bmatrix}$ or $\begin{bmatrix} 0 & \pm 1 \\ \pm 1 & 0 \end{bmatrix}$

$Tr(A) = 0 \Rightarrow$ 6 matrices

$Tr(A) \neq 0 \Rightarrow$ 2 matrices. Here $|A| \neq 0$

47.-49. $g(f(x)) = \begin{cases} x + a + 1 & x < -a \\ (x + a - 1)^2 + b & -a \le x < 0 \\ x^2 + b & 0 \le x < 1 \\ (x - 2)^2 + b & x \ge 1 \end{cases}$

Now for a = 1, b = 0 g(f(x)) is continuous for all $x \in R$

and for a = 1, b = 0 g(f(x)) is non-differentiable at $x = \pm 1$

SECTION – C

50. Let $A_r = (2t, t^2)$

$\therefore$ Slope of $BA_r = \dfrac{t^2 - 1}{2t} = \tan\left(\dfrac{\pi}{2} + \theta_r\right)$

$\therefore \tan(\theta_r) = \dfrac{2t}{1 - t^2} = \tan 2\phi$ (Let)

$\Rightarrow \phi = \dfrac{\theta_r}{2} = \dfrac{r\pi}{4n}$ where $\tan\phi = t$ and $BA_r = \sqrt{(t^2 - 1)^2 + 4t^2} = t^2 + 1 = \sec^2\phi = \sec^2\left(\dfrac{r\pi}{4n}\right)$

$\Rightarrow \lim\limits_{n\to\infty} \dfrac{1}{n}\sum\limits_{r=1}^{n} BA_r = \int\limits_0^1 \sec^2\left(\dfrac{\pi x}{4}\right)dx = \dfrac{4}{\pi}$

51. For f({1, 2, 3}) and f(f{1, 2, 3}) to be disjoint, we must have f(1), f(2), f(3) $\notin$ {1, 2, 3}
If f(1) = f(2) = f(3) = 4 or 5, then one of f(4) and f(5) will have 4 and other have 5 choice. Number of possible function = $2 \times 4 \times 5 = 40$
If two of f(1) and f(2) and f(3) are equal and rest is different then f(4), f(5) $\in$ {1, 2, 3}
$\Rightarrow$ Total number of possible function = $3 \times 2 \times 3 \times 3 = 54$
$\therefore N = 94$

52. The given equation is $(x^{2018} + 1)(1 + x + ... + x^{2016}) = 4034.x^{2017}$

$\Rightarrow \left(x + \dfrac{1}{x^{2017}}\right)\left(1 + x + ... + x^{2016}\right) = 4034$

$\Rightarrow \left(x + \dfrac{1}{x}\right) + \left(x^2 + \dfrac{1}{x^2}\right) + ... + \left(x^{2017} + \dfrac{1}{x^{2017}}\right) = 2.2017$

Here, $x > 0$ therefore L.H.S $\ge 2 \times 2017$ and for equality each term of L.H.S must be equal to 2
$\Rightarrow x = 1$

53. Given equation can be written as $2x^2y(x^2\,dy + 2xy\,dx) + y^2\,dy - x^2\,dx = 0$
$\Rightarrow 2x^2y\,d(x^2y) + y^2\,dy - x^2\,dx = 0$
Integrating, we get $3(x^2y)^2 + y^3 - x^3 = c$

54. Equation of OI is, $2x + y = 4$
Vertex is point of intersection of $2x + y = 4$ and $5x + 2y = 4$
$\therefore A = (-4, 12)$, $AO = 5\sqrt{5}$, $AI = 6\sqrt{5}$
$AG = AI\cos 30^\circ = 3\sqrt{15}$
$\because AG = \dfrac{AE + AB}{2}$
$\therefore AE = 2AG - AB = \sqrt{15}$
$\therefore$ Area of quadrilateral BCDE = area of ΔABC – area of ΔADE
$= \dfrac{\sqrt{3}}{4}\left(AB^2 - AE^2\right) = 90\sqrt{3}$

